THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** metal	Metal properties.
2	**(b)** red litmus blue	Basic.
3	**(b)** a protective oxide layer forms	Al2O3 layer.
4	**(b)** non-metal	Non-metal.
5	**(b)** blue litmus red	Acid.
6	**(b)** it is lustrous	Lustre.
7	**(b)** lead	Lead is a poor conductor among metals.
8	**(b)** milky	Calcium carbonate.
9	**(a)** copper conducts, PVC insulates	Safety.
10	**(c)** graphite	Carbon form.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Silver reacts with sulphur compounds in air (1); Forms black silver sulphide (1).	1+1
12	Magnesium hydroxide as antacid (1); Calcium oxide for neutralising acid (1).	1+1
13	Iron is strong and hard (1); Sodium is soft and very reactive (1).	1+1
14	It forms a protective oxide layer (1); The layer protects the metal inside (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No reaction (1); Copper is less reactive (1); It cannot displace iron (1).	1+1+1
16	Coating is basic copper carbonate or oxide (1); Acid in lemon reacts (1); Surface becomes shiny (1).	1+1+1
17	Heat spreads (1); Food cooks quickly (1); Handles remain cool (1).	1+1+1
	OR: Take three test tubes (1); Add metal strips (1); Zinc bubbles fast, iron slowly, copper not at all (1).	1+1+1
18	Iron: hard, magnetic, conducts (1); Sulphur: brittle, yellow, insulator (1); Different type (1).	1+1+1
	OR: MgO: magnesium oxide (1); SO_2: sulphur dioxide (1); Formulae show atoms (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Collect samples (1); Observe lustre (1); Test conduction (1); Strike and listen (1); Table and conclusion (1).	1+1+1+1+1
	OR: Hold with tongs (1); Do not look directly (1); Collect the ash (1); Dissolve in water (1); Test with litmus (1).	1+1+1+1+1
20	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $S + O_2 \longrightarrow SO_2$ (1); Dissolve in water (1); MgO basic, SO2 acidic (1); Litmus test (1).	1+1+1+1+1
	OR: Iron in blood (1); Oxygen for breathing (1); Carbon in food (1); Metals in machines (1); Nitrogen in fertilisers (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** hydrogen	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$.
2	**(b)** gold is less reactive	Reactivity differs.
3	**(b)** carbon and sulphur	C and S.
4	**(c)** copper	Copper is below hydrogen.
5	**(c)** gold is unreactive	Does not corrode.
6	**(d)** copper	Ductile conductor.
7	**(d)** sodium	Alkali metal.
8	**(d)** iron conducts heat	So that it is safe.
9	**(d)** copper and rubber	Contrast.
10	**(a)** iron	More reactive.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reacts with moist air (1); Forms basic copper carbonate (1).	1+1
12	It is lustrous (1); Most non-metals are dull (1).	1+1
13	Burns with a pop sound (1); Bring a burning splinter near the gas (1).	1+1
14	Iron (1); It displaces copper from copper sulphate (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Aluminium: malleable (1); Copper: ductile conductor (1); Metal: sonorous (1).	1+1+1
16	A new compound forms (1); No longer attracted by magnet (1); Properties differ (1).	1+1+1
17	Sulphur is a poor conductor (1); It is brittle and breaks when bent (1); Copper is ductile and conducts well (1).	1+1+1
	OR: Gives carbon dioxide (1); Dissolves to form carbonic acid (1); Turns blue litmus red (1).	1+1+1
18	Improve strength (1); Resist rust (1); Examples: steel, brass (1).	1+1+1
	OR: Copper does not react (1); It is less reactive than hydrogen (1); No bubbles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Aluminium foil: malleable (1); Copper wire: ductile conductor (1); Iron girders: strong (1); Gold: lustrous, unreactive (1); Mercury: liquid (1).	1+1+1+1+1
	OR: Burn carbon (1); Collect gas in a jar (1); Add water and shake (1); Add blue litmus (1); It turns red (1).	1+1+1+1+1
20	Fe + CuSO4: displaces (1); Zn + CuSO4: displaces (1); Cu + FeSO4: no reaction (1); Zinc > iron > copper (1); Order from reactions (1).	1+1+1+1+1
	OR: Magnesium fastest (1); Zinc moderate (1); Iron slow (1); Copper none (1); Mg > Zn > Fe > Cu (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** they ring clearly	Sonorous.
2	**(b)** a metal	Sonorous.
3	**(a)** silver	Less reactive.
4	**(c)** they are hard and malleable	Can be stamped.
5	**(a)** they are malleable	Malleable.
6	**(c)** mercury	Liquid at room temperature.
7	**(c)** all non-metals are gases	Non-metals are solids, liquids, gases.
8	**(b)** sulphur	Brittle non-metal.
9	**(a)** sodium and potassium	Highly reactive.
10	**(b)** malleability	Sheets.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Metals conduct electricity (1); They carry the charge to the ground (1).	1+1
12	They are brittle (1); They are poor conductors (1).	1+1
13	$S + O_2 \longrightarrow SO_2$ (1); Sulphur dioxide (1).	1+1
14	$C + O_2 \longrightarrow CO_2$ (1); Carbon dioxide formed (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dissolve each in water (1); Test with litmus (1); Magnesium oxide is basic, sulphur dioxide is acidic (1).	1+1+1
16	Copper reacts with acidic food (1); Tin coating prevents it (1); Food stays safe (1).	1+1+1
17	Metal oxides basic (1); Non-metal oxides acidic (1); Litmus test (1).	1+1+1
	OR: Very reactive (1); Catches fire in moisture (1); Soft metal (1).	1+1+1
18	CO2 does not support burning (1); Oxygen is limited (1); Used in fire extinguishers (1).	1+1+1
	OR: Save ores (1); Save energy (1); Reduce waste (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mercury liquid (1); Sodium soft (1); Graphite conducts (1); Iodine lustrous (1); Most properties decide (1).	1+1+1+1+1
	OR: Painting (1); Oiling (1); Galvanising (1); Alloying (1); Plating (1).	1+1+1+1+1
20	Iron rusts (1); Silver tarnishes (1); Copper turns green (1); Painting (1); Galvanising (1).	1+1+1+1+1
	OR: Iron Pillar does not rust easily (1); Skilled ancient metallurgy (1); Phosphorus-rich iron (1); Dry climate helps (1); Heritage (1).	1+1+1+1+1

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